

BONUS TASK

Power Consumption Analyzer

BACKGROUND

In electrical power systems, the instantaneous power consumed by a load is given by:

$$P = V \times I \quad \text{(Watts)}$$

$$E = P \times t \quad \text{(Wh, where } t = 1 \text{ hour)}$$

Recording voltage and current every hour over a full day allows engineers to profile the daily energy consumption of a system. This task asks you to automate that analysis in Java.

REQUIREMENTS

- 01

Write a static method with the following signature that takes voltage and current as parameters and returns the computed power:

```
public static double  
calculatePower(double voltage,  
double current)
```
- 02

In main, declare three arrays of size 24 to store the voltage readings, current readings, and computed power values for each hour of the day.
- 03

Use a loop to read voltage and current from the user for all 24 hours. Call calculatePower to populate the power array.
- 04

After all input is collected, compute and print a full hourly report showing voltage, current, and power for each hour. Any hour where power exceeds 1000 W must be flagged as High Consumption.
- 05

Compute and print a summary that includes: total energy consumed over 24 hours (Wh), average power (W), the peak hour and its power value, and the total count of high-consumption hours.

EXPECTED OUTPUT FORMAT

```
--- Hourly Power Report ---  
Hour  1 : V =  220.0 V   I =  3.5 A   P =   770.0 W  
Hour  2 : V =  220.0 V   I =  5.2 A   P =  1144.0 W  << High Consumption  
Hour  3 : V =  215.0 V   I =  2.1 A   P =   451.5 W  
      . . .  
-----  
Total Energy Consumed : 18500.0 Wh  
Average Power         :   770.8 W  
Peak Hour              : Hour 2  - 1144.0 W  
High Consumption Hours : 6 hour(s)
```